

T3A1 - Weight Functions and Sample Variance

Setup

$S = \int_0^1 x^2 dx = 1/3$. Sample points x_i from a density $W(x)$ with $\int_0^1 W = 1$, then estimate any $A(x)$ as an average:

$$\langle A \rangle_W = \frac{1}{N} \sum_i A(x_i)$$

Pick $A(x) = x^2/W(x)$ so this always equals to S

a) $W(x) = 1$ (uniform)

$$A(x) = x^2.$$

$$\langle A \rangle_W = \int_0^1 x^2 dx = \frac{1}{3}$$

The result matches S

b) Variance (uniform)

$$\langle A^2 \rangle = \frac{15}{45}, \quad \sigma^2 = \frac{15}{45} - \left(\frac{1}{3}\right)^2 = \frac{14}{45}$$

c) $W(x) = 2x$

$$A(x) = x/2.$$

$$\sigma^2 = \frac{18}{72} - \left(\frac{1}{3}\right)^2 = \frac{1}{72}$$

~6× less variance than uniform

d) $W(x) = 3x^2$

$$A(x) = 1/3 \text{ (constant)} \rightarrow \sigma^2 = 0.$$

Explanation: every point gives $A(x_i)=1/3$, so one sample nails it because $W(x)=3x^2$ equals $f(x)/S$, and S is exactly what we're solving for. The ideal density is proportional to the integrand, but you need S upfront to build it, so it's a theoretical limit, not something you can use.

Python implementations

To draw x_i from a nonuniform $W(x)$, invert its CDF:

$$x = F^{-1}(u), \quad \text{where } F(x) = \int_0^x W(t) dt, \quad \text{where } u \sim \text{Uniform}(0,1)$$

For $W(x) = 2x$, $F(x) = x^2$, so $x = \sqrt{u}$; for $W(x) = 3x^2$, $F(x) = x^3$, so $x = u^{1/3}$.

```
import numpy as np

def mcS_uni(N, rng):
    """W(x)=1; A(x)=x^2"""
    x = rng.uniform(0, 1, N)
    return np.mean(x**2) # just average x^2

def mcS_bias(N, rng):
    """W(x)=2x; A(x)=x/2"""
    u = rng.uniform(0, 1, N)
    x = np.sqrt(u) # bend the uniform draw so it favors x near 1
    return np.mean(x / 2) # A(x)=x/2 this time

def mcS_optimal(N, rng):
    """W(x)=3x^2; A(x)=1/3"""
    u = rng.uniform(0, 1, N)
    x = u ** (1/3) # even more skewed towards x=1
    return np.full(N, 1/3).mean() # A is constant so this is just 1/3
```

All 3 converge to $1/3$, but the scatter around it shrinks from `mcS_uni` to `mcS_bias` to `mcS_optimal` (exact for any N , matching the result above).